Week 16: Three-color triangles

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

The answer keys, proofs, held hints and witness figures are in the print guide, week-16-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py with route-note.json.

Problems and routes

Problem	K–1	2–3	4–5	route
One rainbow triangle in each of four places	P1			first
Fewest and most rainbow triangles on the three-step board		P1	P1	first
Can every rainbow triangle be avoided?	P2			first
Doors and routes on a printed filling		P2	P2	first
Most rainbow triangles on the three-step board	P3			first
Every filling of one little triangle		P3	P3	return
Most rainbow triangles on the fan board	P4			return
Doors along a row of dots		P4	P4	return
One rainbow triangle in every place on the four-step board	P5			return
Exactly three rainbow triangles, and the outside routes		P5		return
When the bottom side allows Y	P6			return
Must the number of rainbow triangles be odd?			P5	return
Can the fan board have none?		P6		return
One starred side dot may use any letter			P6	return

First route. K–1 P1–P3; Grades 2–3 P1–P2; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P6; Grades 2–3 P3–P6; Grades 4–5 P3–P6.

Key, one entry per problem

One rainbow triangle in each of four places

K–1 P1 · first route

Idea. On the two-step board every legal filling has exactly one triangle with R, B and Y, and it can sit in any of the four places.

Fewest and most rainbow triangles on the three-step board

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. On the three-step board only 1, 3 and 5 triangles with R, B and Y occur.

Can every rainbow triangle be avoided?

K–1 P2 · first route

Idea. With the side rules kept, no filling avoids a triangle with R, B and Y.

Doors and routes on a printed filling

Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. Routes through R-B doors pair outside doors two at a time or end in a triangle with R, B and Y; an odd number of outside doors leaves one unpaired.

Most rainbow triangles on the three-step board

K–1 P3 · first route

Idea. The three-step board holds at most five triangles with R, B and Y.

Every filling of one little triangle

Grades 2–3 P3 · Grades 4–5 P3 · return route · wording differs by band

Idea. Of the ten fillings of one triangle up to turns and flips, only RBY has one door; RRB and RBB have two; none has three.

Most rainbow triangles on the fan board

K–1 P4 · return route

Idea. With its printed letters kept, the fan board holds at most seven triangles with R, B and Y.

Doors along a row of dots

Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. A row from R to B has an odd number of doors and a row from R to R an even number, whatever its length.

One rainbow triangle in every place on the four-step board

K–1 P5 · return route

Idea. Each of the 16 little triangles of the four-step board can be the only one with R, B and Y.

Exactly three rainbow triangles, and the outside routes

Grades 2–3 P5 · return route

Idea. Exactly three is possible on the four-step board, and the outside routes can never all end outside because the outside doors are odd in number.

When the bottom side allows Y

K–1 P6 · return route

Idea. Once the bottom side may use Y, fillings with no triangle with R, B and Y exist.

Must the number of rainbow triangles be odd?

Grades 4–5 P5 · return route

Idea. Sperner's lemma: on every edge-to-edge triangulation with the side rules, the number of triangles with R, B and Y is odd ($T + 2U = b + 2I$).

Can the fan board have none?

Grades 2–3 P6 · return route

Idea. No filling allowed by the side rules has zero triangles with R, B and Y, on any board.

One starred side dot may use any letter

Grades 4–5 P6 · return route

Idea. Relaxing one side dot breaks the hypothesis: a filling with no triangle with R, B and Y exists.